

End Semester Examination – winter- 2022

Course: B. Pharmacy

Sem: 6th

Subject Name: Medicinal Chemistry-III

Subject Code: BP601T

Max Marks: 75

Date: 07/02/2023

Duration: 3 Hr

Instructions:

- 1. All questions are compulsory**
 - 2. Draw diagrams / figures wherever necessary**
 - 3. Figures to right indicate full marks**
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Q. 1. Objective Type questions (Answer all the questions) (10* 2) =20

- Write the structure and uses of Penicillin-G.
- Give the synthesis of Isoniazid.
- List out important anti-viral agents. Draw the structure of acyclovir.
- Write the structure of any two anti-malarial drugs.
- What are antiprotozoal agents? Write the structure of metronidazole.
- What are anthelmintics? Give examples.
- Give the structure and IUPAC name of Dapsone.
- Write the synthesis of para amino salicylic acid.
- Define anti-tubercular agents? Name the causative organism for tuberculosis.
- Enlist the parameters of QSAR.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- Define and Classify antimalarial agents with examples. Give the mechanism of action and outline the synthesis of chloroquine.
- What are anthelmintics? Classify with suitable examples. Outline the synthesis and mechanism of action of Mebendazole.
- What are antibiotics? Classify with examples. Discuss the SAR & MOA of tetracyclines.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- What are aminoglycosides? Write the mechanism and SAR of aminoglycoside antibiotics.
- Write a note on urinary tract anti-infective agents.
- What are Sulphonamides? Explain their SAR.
- Write a note on combinatorial chemistry and its applications.
- Define and classify anti-tubercular agents with examples. Write the MOA of isoniazid.
- Add a note on synthetic antifungal agents. Give the synthesis of Tolnaftate.
- Write a note on prodrug concept and applications of prodrugs design.